Leveraging Sequence-Structure Integration with Bandit-Based Sampling for Receptor-Binding Protein Identification

1. SUPPLEMENTARY NOTE

S1.1 Sequence Branch

The sequence branch uses ESM2 (`esm2_t33_650M_UR50D`) [1], a pre-trained protein language model, to extract representations from input protein sequences. Given an input sequence, ESM2 produces a high-dimensional embedding for each residue position:

$$\mathbf{X}_{1d} = \text{ESM2}(\text{sequence}) \in \mathbb{R}^{B \times L_1 \times D_1}, \tag{S1}$$

where B denotes the batch size, L_1 denotes the sequence length, and D_1 denotes the per-token embedding dimension. The embeddings are then processed by a Transformer encoder to obtain token-level hidden representations $\mathbf{H}_{1d} \in \mathbb{R}^{B \times L_1 \times d}$, where d denotes the hidden dimension. Mean pooling over the sequence dimension yields a fixed-length, sample-level representation:

$$\mathbf{Z}_{1d} = \frac{1}{L_1} \sum_{i=1}^{L_1} \mathbf{H}_{1d}^{(i)} \in \mathbb{R}^{B \times d}. \tag{S2}$$

S1.2 Structure Branch

The structure branch encodes structure-derived features obtained by first predicting three-dimensional structures with ColabFold [2] and then extracting structure-aware representations using Saprot (`SaProt_650M_AF2`) [3]. Specifically, ColabFold generates a predicted 3D structure for each input sequence, which is then processed by Saprot to produce residue-level structural features:

$$\mathbf{X}_{3d} = \text{Saprot}(\text{ColabFold}(\text{sequence})) \in \mathbb{R}^{B \times L_3 \times D_3}, \tag{S3}$$

where L_3 denotes the number of residues in the predicted structure and D_3 denotes the per-residue feature dimension. These features are fed into a Transformer encoder to obtain token-level hidden representations $\mathbf{H}_{3d} \in \mathbb{R}^{B \times L_3 \times d}$, where d denotes the hidden dimension. Mean pooling over residues produces the sample-level representation $\mathbf{Z}_{3d}$:

$$\mathbf{Z}_{3d} = \frac{1}{L_3} \sum_{i=1}^{L_3} \mathbf{H}_{3d}^{(i)} \in \mathbb{R}^{B \times d}. \tag{S4}$$

S1.3 Experimental Details

The batch size B is set to 32, and the feature dimension of the Transformer is fixed to $d = 128$. The sequence length L_1 is truncated to 1024, and the embedding dimension D_1 is 1280. For the structure branch, the L_3 truncated to 512 and the embedding dimension D_3 set to 1280. In the AEFM, the rank parameter r is set to 64. All experiments are conducted on a single A100 under fixed random seeds to ensure reproducibility.

To control computational cost while maintaining stable utility estimation, training uses a fixed-size active negative pool. In the initial training phase, which uses only sequence data, 20,000 negative samples are uniformly drawn to form the first active pool. Every two epochs, the pool is refreshed by retaining the 10,000 highest-utility negatives from the previous window, while the remaining 10,000 negatives are replaced with newly drawn samples from the unseen negative set. This retain-refresh scheme enforces a controlled exploration rate, reducing the risk of overfitting to a static subset. Once training with the sequence data is completed, we use the last retained 20,000 negatives to predict the 3D structures of the proteins. These predicted 3D structures are

then incorporated into the training, moving to a Sequence Combined with Structure setup. At this stage, the negative sample pool continues to follow the similar retain-refresh scheme, where the 1,000 highest-utility negatives from the previous window are retained, and the remaining 1,000 negatives are refreshed with new samples from the unseen negative set. The UCB exploration coefficient is set to the canonical value $C = \sqrt{2}$, which provides a scale-consistent uncertainty bonus without requiring dataset-specific tuning.

S1.4 Description of the Metrics

Mathematically, AUC is defined as the integral of the area beneath the Receiver Operating Characteristic (ROC) curve, and its calculation follows a well-established formula as follows:

$$\text{AUC} = \int_0^1 TPR \cdot FPR \; d(FPR) \tag{S5}$$

where

$$TPR = \frac{TP}{TP + FN}, \qquad FPR = \frac{FP}{FP + TN} \tag{S6}$$

The True Positive Rate (TPR) reflects model's ability to correctly identify RBPs, while the False Positive Rate (FPR) measures the frequency of errors in which non-RBPs are misclassified as RBPs. The ROC curve itself plots the relationship between TPR and FPR across all possible classification thresholds, and AUC provides a single quantitative value that summarizes the overall quality of this curve. The value of AUC lies within the interval [0, 1] with values closer to 1 indicating a stronger ability to distinguish between RBPs and non-RBPs. A high AUC score means that the model can, with high probability, rank a randomly selected true RBP ahead of a randomly selected non-RBP.

S1.5 Parameter analysis

S1.5.1 Analysis of Resampling Interval in Bandit-Based Negative Selection

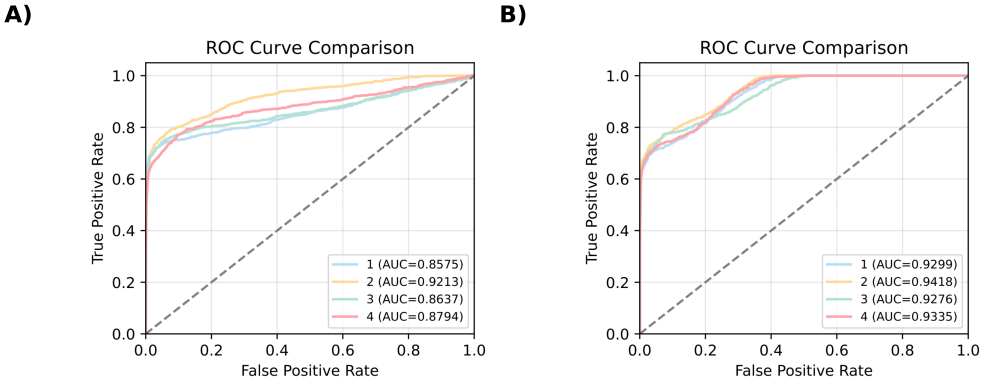

Fig. S1. ROC curves of different resampling interval. A) Sequence-only model; B) Sequence combined with structural information.

We analyzed the sensitivity of the resampling interval, a key hyperparameter in the bandit-based negative sampling framework, using ROC curves as the evaluation metric. Fig S1 compare the ROC curves obtained under different resampling intervals for both the sequence-only model and the sequence combined with structure model.

For the sequence-only model, the overall performance varies across different intervals. Among them, interval 2 achieves the highest area under the ROC curve (AUROC = 0.9213), indicating the best overall discrimination capability between positive and negative samples. In comparison, interval 1 results in a lower AUROC (0.8575), while intervals 3 and 4 achieve AUROC values of 0.8637 and 0.8794, respectively. These results suggest that moderate policy update frequency helps the bandit framework better identify informative negative samples.

For the model that combines sequence and structural information, all settings show improved ROC performance compared with the sequence-only model. Interval 2 again achieves the best performance with the highest AUROC (0.9418). The other intervals yield slightly lower but still

competitive results, with AUROC values of 0.9299, 0.9276, and 0.9335 for intervals 1, 3, and 4, respectively. The consistent advantage of interval 2 across both settings indicates that this interval provides a balanced trade-off between exploration and exploitation in the bandit-based negative sampling strategy.

Overall, the ROC analysis demonstrates that an intermediate resampling interval yields the most stable and discriminative performance. Therefore, we adopt a resampling interval of 2 in all subsequent experiments.

S1.5.2 Analysis of Retained and Newly Sampled Negatives

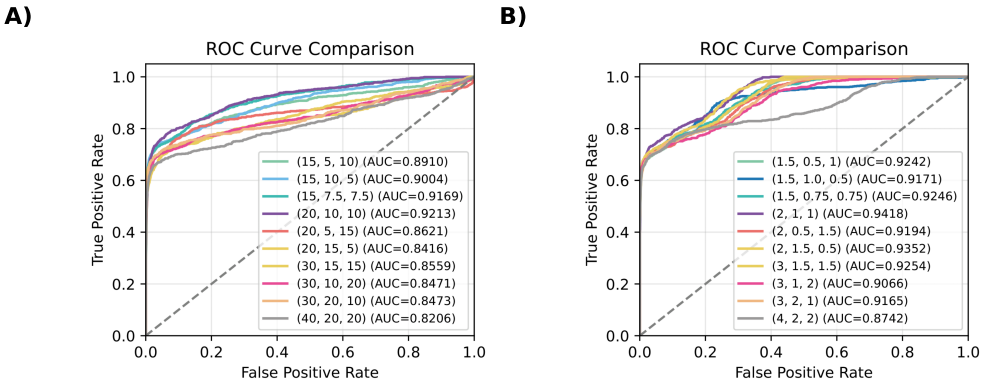

Fig. S2. ROC curves of different sample retention and replacement ratio. The tuple (N, R, S) represents the total number of negatives, retained negatives, and newly sampled negatives, respectively. A) Sequence-only model; B) Sequence combined with structural information.

We further analyzed the effect of the retention–replacement ratio in the negative resampling process using ROC curves. Fig. S2 compares the ROC curves obtained under different retention–replacement configurations for both the sequence-only model and the sequence combined with structural information.

For the sequence-only model, the ROC analysis shows that configurations with a balanced ratio between retained negatives and newly sampled negatives generally achieve better discrimination performance. In particular, the configuration $(20, 10, 10)$ yields the highest AUROC value (0.9213), indicating the best overall ability to distinguish positive and negative samples. Other relatively balanced configurations, such as $(15, 7.5, 7.5)$ and $(15, 10, 5)$, also achieve competitive performance with AUROC values of 0.9169 and 0.9004, respectively. In contrast, highly imbalanced strategies, such as $(40, 20, 20)$, result in noticeably lower performance (AUROC = 0.8206), suggesting that excessively large negative pools or disproportionate retention may reduce the effectiveness of hard negative mining.

A similar trend is observed for the model that integrates sequence and structural information. Among all tested configurations, the balanced setting $(2, 1, 1)$ achieves the highest AUROC value (0.9418). Other configurations with relatively balanced retention and replacement ratios, such as $(1.5, 0.75, 0.75)$ and $(1.5, 0.5, 1)$, also show strong performance with AUROC values of 0.9246 and 0.9242, respectively. In contrast, imbalanced configurations such as $(4, 2, 2)$ lead to lower AUROC values (0.8742), indicating reduced discriminative capability.

Overall, the ROC analysis demonstrates that maintaining a balanced proportion between retained hard negatives and newly sampled negatives provides the most stable and effective performance. This balance enables the model to simultaneously exploit informative hard negatives and introduce sufficient diversity through newly sampled examples. Based on these results, we adopt an equal retention–replacement strategy in subsequent experiments.

S1.6 Experimental Configuration for Phage-Host Interaction Prediction

The interaction prediction is formulated as a binary classification task, where the model predicts the presence (labeled as 1) or absence (labeled as 0) of an interaction between a phage and a host. The initial dataset comprised 22 phages and 28 *Vibrio* genomes, encompassing 142 experimentally confirmed interactions. Subsequently, we filtered out entities with zero recorded interactions,

resulting in a refined set of 19 phages and 25 *Vibrio* genomes. The final dataset consists of 475 pairs, comprising 142 interacting pairs and 333 non-interacting pairs.

The input features for each pair aggregate information from both phage and bacterial components. We utilized the **Observed** and **Predicted** sets of RBPs to represent phage information. For the bacterial representation, O-antigen biosynthesis-related proteins are employed, as they determine the structure of the O-antigen, which serves as the primary receptor for bacteriophage tail fiber binding. All RBP sequences from a phage and O-antigen proteins from a bacterium are embedded using the ESM-2 transformer model (*esm2_t33_650M_UR50D*). These embeddings are then averaged to produce a single feature vector per entity. Finally, the phage and bacterial feature vectors are concatenated into a 2560-dimensional vector to represent each interaction pair.

We implemented a multilayer perceptron (MLP) with a single hidden layer for interaction prediction. The model was trained for 10 epochs using the Adam optimizer with a learning rate of 1×10^{-4}. To ensure robust evaluation, we employed a 10-fold stratified cross-validation scheme. The final results are shown in Fig. S3.

S1.7 Experimental Configuration for RPBs and Outer Membrane Proteins Docking

To validate the biological relevance of the host predictions, we employed AlphaFold3 to estimate the docking quality (measured by interface pLDDT) between phage receptor-binding proteins (RBPs) and outer membrane proteins from the host genomes. Due to the high computational demands of AlphaFold3, we selected a subset of five phages for this comparison. As shown in Fig. S2, these five phages (202005_zK_SDZB, 6, $P15$, $P63$, $P72$) exhibit the most substantial differences in RBP prediction results (**Observed** vs. **Predicted**). We inputted the non-overlapping RBPs of these phages and their corresponding host outer membrane proteins into AlphaFold3, calculated the interface pLDDT scores, and visualized the distributions as a box plot (Fig. S4).

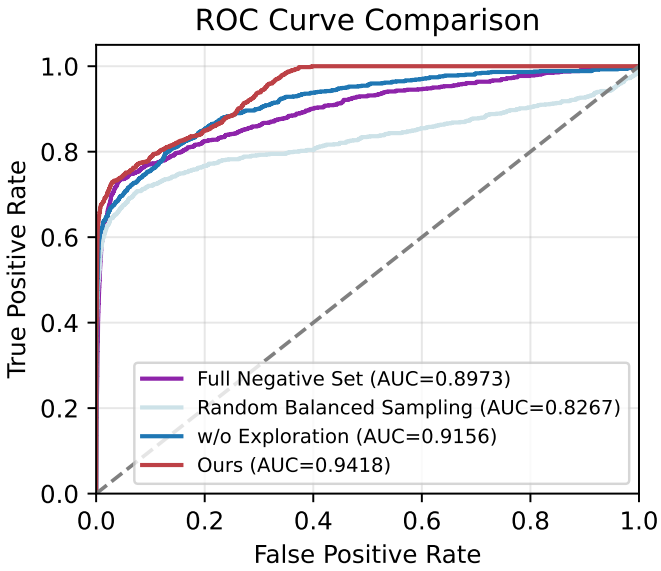

Fig. S3. ROC curves of different negative sampling strategies evaluated on the constructed test set only using sequence features.

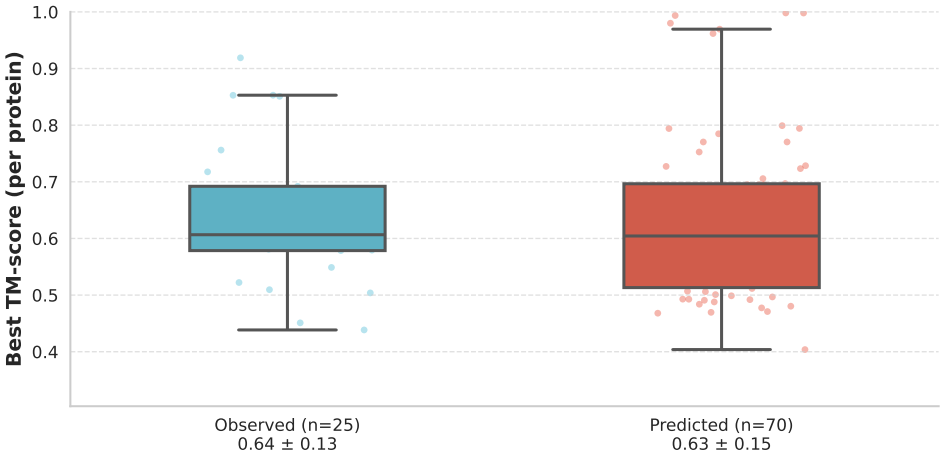

Fig. S4. TM-score comparison between predicted RBPs (**Predicted**) and manually selected RBPs (**Observed**) in the *Vibrio* dataset. The predicted RBPs show larger TM-scores range, possibly reflecting discovered divergent RBP families.

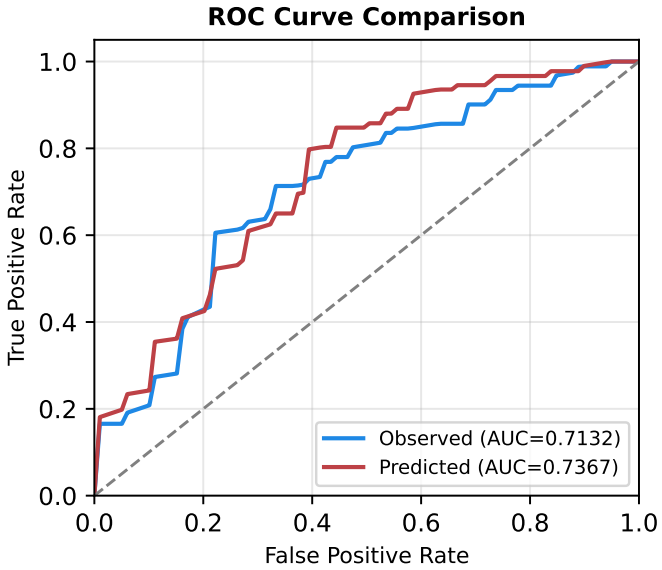

Fig. S5. ROC curves of phage-host interaction prediction on the *Vibrio* dataset. Performance is evaluated for the **Observed** set (blue, AUC=0.7132) and the **Predicted** set (red, AUC=0.7367), with their mutual intersection excluded to isolate unique candidates. The higher AUROC for the Predicted candidates demonstrates that RBPs identified by SeekRBP capture robust functional signals missed by experimental annotations.

REFERENCES

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